C++ Notes: Working With Numbers

(Summary from the Numbers section of the course)

Author: Ba Tung Le

Learning Objectives

After this section, you will be able to:

- Work with whole numbers (`int`) and decimal numbers (`double`)
- Use arithmetic operators: `+`, `-`, `*`, `/`
- Use the modulus operator `%` (remainder)
- Understand order of operations and parentheses
- Use increment/decrement and shorthand operators (`++`, `+=`, etc.)
- Use basic math functions from `<cmath>`

Core Idea

Core idea

Numbers are one of the most common types of data in C++.

- `int` = whole numbers (no decimal point)
- `double` = decimal numbers (more precision than `float`)

Printing Numbers and Basic Math

Printing a number

```
cout << 14;
```

Output:

```
14
```

Addition, subtraction, multiplication, division

```
cout << 5 + 7 << endl;
cout << 10 - 3 << endl;
cout << 5 * 7 << endl;
cout << 10 / 2 << endl;
```

Output:

```
12
7
35
5
```

Tip

You can write math directly inside `cout`. C++ will evaluate the expression and print the result.

Modulus Operator (%)

```
cout << 10 % 3;
```

Output:

```
1
```

Modulus

`a % b` returns the remainder of `a / b`. Example: `10 / 3 = 3 remainder 1`, so `10 % 3 = 1`.

Order of Operations

Without parentheses:

```
cout << 4 + 5 * 10;
```

Output:

```
54
```

With parentheses:

```
cout << (4 + 5) * 10;
```

Output:

```
90
```

Reminder

Multiplication and division happen before addition and subtraction. Use parentheses when you want a different order.

Numbers in Variables

```
int wNum = 5;
double dNum = 5.5;

cout << wNum << endl;
cout << dNum;
```

Output:

```
5
5.5
```

Increment / Decrement

Before:

```
int wNum = 5;
cout << wNum;
```

Output:

```
5
```

After `wNum++`:

```
int wNum = 5;
wNum++;
cout << wNum;
```

Output:

```
6
```

After `wNum--`:

```
int wNum = 5;
wNum--;
cout << wNum;
```

Output:

```
4
```

Shorthand Operators

```
int wNum = 5;

wNum += 80; // wNum = wNum + 80
cout << wNum;
```

Output:

```
85
```

```
int x = 10;

x *= 3; // x = x * 3
cout << x;
```

Output:

```
30
```

Integer vs Decimal Division

Integer division (truncates decimals)

```
cout << 10 / 3;
```

Output:

```
3
```

Important

When both operands are integers, C++ returns an integer result. The decimal part is discarded.

Decimal division (keeps decimals)

```
cout << 10.0 / 3;
```

Output:

```
3.33333
```

Mixing decimal and integer

```
cout << 5.5 + 9;
```

Output:

```
14.5
```

Rule

If one side is decimal (`double/float`), the result is decimal.

Math Functions (`<cmath>`)

Setup

To use math functions, include:

```
#include <cmath>
```

pow (power)

```
cout << pow(2, 5);
```

Output:

```
32
```

sqrt (square root)

```
cout << sqrt(36);
```

Output:

```
6
```

round (normal rounding)

```
cout << round(4.3) << endl;
cout << round(4.6);
```

Output:

```
4
5
```

ceil and floor

```
cout << ceil(4.1) << endl;
cout << floor(4.8);
```

Output:

```
5
4
```

fmax and fmin

```
cout << fmax(3, 10) << endl;
cout << fmin(3, 10);
```

Output:

```
10
3
```

Common Mistakes

Common mistakes

- Expecting `10 / 3` to output `3.33...` (it outputs `3`)
- Using `%` with decimal numbers (modulus is for integers)
- Forgetting parentheses when you need a different order
- Forgetting to include `<cmath>` when using math functions

Ba Tung Le
Email: tombn2607@gmail.com